MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON, AMPLIFIER, PULSED

TYPE DOD-034

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Klystron, Pulsed, Amplifier, 1.6 kW, 9.155-9.245 GHz, 5 integral cavities, waveguide input and output, individual tuners, permanent magnet focused, conduction cooled

ABSOLUTE RATINGS: (NOTE 1)

Parameter:	Ef	If (Surge)	Eb (note 2)	Eb	ib	Pi	Ec	ic (See note 3)
Unit:	V	A	kVdc	kVdc	ma	W	kVdc	ua
Maximum:	6	5	13.5	9.5	850	125	-6	500
Minimum:	---	---	---	---	---	---	---	

Parameter:	Du	tp	tk	VSWR	Ambient Field (See note 4)
Unit:		µSEC	Minutes		H
Maximum:	.025	1	---	3:1	5
Minimum:	---	---	2	---	---

Parameter:	Temperature Operating (See note 5)	Altitude Operating
Unit:	C	ft
Maximum:	225	10,000
Minimum:	-40	---

PHYSICAL CHARACTERISTICS:

Mounting Position:	Any
Dimensions:	See Figure 1
Focusing:	Permanent Magnet
Base:	Molded, Color Coded Leads
Cooling:	Conduction
Weight:	9 lbs. Maximum
Tuning:	See Part Nos. 10549169 and 10548235 (Note 7)
RF Flanges:	Mates with UG - 40A/U, or equivalent

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TEST CONDITIONS

Parameter:	Ef	Eb	Ec	F	Load		Du (Video)
Unit:	V	kVdc	kVdc	GHz	VSWR	Mod	Pulse Width/PRT
	5.25	9.0	-4.25	9.205	1.05:1 (Max)		.02 (Max)

Parameter:	TP (RF)	PRF (Rectangular)
Unit:	μ sec	KHz
	.3 ± .01	20 ± .02

GENERAL:

Qualification - Not Required

Holding Period - t - 48 hours (min)

Storage - Note 4

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MIN	MAX	UNIT
	Quality conformance inspection, part 1 (Q.C.I.1)	6					
---	Visual - Mechanical Examination	---	Per Outline Drawing	---	---	---	---
4289	Heater Current	---		If	2.0	2.6	Adc
---	Control Grid - Cathode Leakage	---	Ec = -4.75 kVdc Ef = 5.25V Eb = 9.0kVdc	Ihg	---	100	uA
---	Power Output (1) (Saturated)	3, 22, 23	F = 9.245GHz	Po	1.6	---	kw
---	Power Output Variation	8	F = 9.200 ± 0.045GHz	ΔPo	---	±.5	db
---	Linearity	7, 21	F = 9.205 & 9.245GHz	ΔG	---	0.6	db
---	Gain (Saturated)	3, 22	F = 9.205 & 9.245GHz	G	63	---	db
---	Gain (Linear)	3, 7	F = 9.205 & 9.245GHz Po = 1KW	G	67	---	db

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MIN	MAX	UNIT
	Quality conformance inspection, part 1 (Cont.)						
---	Beam Current	3	Pd = 0	Ib	---	.74	A
---	Cathode, Grid-Cathode voltage Stand-off		Ec = -7.0kVdc Eb = 13kVdc		No Breakdown		
	Quality conformance inspection, part 3						
---	Altitude	9	Po(1)	Po	1.6	---	kw
---	Spurious Output	11	Po(1)	---	---	---	---
4260	Noise Figure	---	Po(1)	NF	---	38	db
4303	Warm-up Time	12	---	Po	1.02	---	kw
---	Tunable Frequencies	7	Po(2)	---	---	---	---
1011	Humidity	---	Po(1)	---	---	---	---
---	Bandwidth	20	Po(2)	---	5	---	MHz
---	Life Test	13	Po(2)	t	2000	---	hrs
	Life -Test End Point						
---	Power Output	3, 22	Po(2)	---	.8	---	kw
---	Control Grid-Cathode Leakage		Ec=4.25KVdc Ef=5.25V Eb=9.0KVdc	Ihg	---	200	uAdc
	Quality conformance inspection, part 3 (Cont.)						
	Periodic-Check Test						

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	MIN	LIMITS MAX	LIMITS UNIT
1026	Temperature Operating	14	Po(1)	Po	1.6	---	kw
1031	Vibration Sweep Frequency	15	Po = 1 kw	---	---	---	---
---	Shock, Operating	16	Po(2) Peak Acceleration = 15G t = 11 ± 1 milliseconds	---	---	---	---
---	Shock Non-Operating	17	No Voltages Peak Acceleration = 75G t = 11 ± 1 milliseconds				
1331	Capacitance Direct Interelectrode (Grid to Cathode)	18	Leads Bundled	Cgk	---	23	pf
4223	Mechanical Tuning Fatigue	19	---	---	---	---	---
---	Heater Aging	---	5 min on, 5 min off, 10 cycles	---	No Open Heaters	---	---

NOTES:

1. The absolute ratings are values which are not to be exceeded under any service conditions. These ratings are limiting values beyond which the serviceability of any individual tube may be impaired. In order not to exceed absolute ratings, the designer has the responsibility of determining an average design value for each rating relative to the absolute value of that rating by a safety factor so that the absolute values will never be exceeded under any usual conditions of supply-voltage variation, load variation, or manufacturing variations in the equipment itself. It does not necessarily follow that combinations of absolute maximum ratings can be attained simultaneously.

2. Non-conducting value.

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NOTES: (Cont.)

3. The pulse voltage applied to the control electrode should be equal to, but should not exceed, the control electrode bias voltage for optimum tube performance. The tube shall not be degraded, however, with the application of a peak positive pulse overshoot of 150V with recovery to a 50.0V overshoot point occuring at less than 20% of the pulse duration.

4. An ambient field of 5H or less shall not result in degraded performance. Ferromagnetic materials shall be kept at least 6 inches away from the tube and other magnets shall be kept at least 12 inches distant.

5. The temperature shall be measured on the collector core.

6. First Article tests shall be in accordance with the contract or purchase order with no failures accepted. Each tube shall be tested in accordance with this specification on a 100% basis with no failures accepted. As a control sample, one tube shall be selected at random by a Government representative from each 50 tubes produced and shall meet all the requirements of First Article except life test with no failures accepted.

7. Coarse tuning shall result in being able to tune-in any of the frequencies listed ($\pm$ 1MHz) below in less than one minute, using tuning tools noted below, and obtaining at least 100W.

LOW FREQUENCY LIMIT:	9.155 GHz
CHANNEL 1:	9.205 GHz
CHANNEL 2:	9.213 GHz
CHANNEL 3:	9.221 GHz
CHANNEL 4:	9.229 GHz
CHANNEL 5:	9.237 GHz
CHANNEL 6:	9.245 GHz

Fine tuning shall result in obtaining a nominal power-out of 1.0 kw.

NOTE: Tuning tools per frequency range 9.2 to 9.25 GHz - 10549169
9.15 to 9.25 GHz - 10548235

8. The output power shall not vary more than $\pm$ 1/2 db over the frequency range 9.200 $\pm$ 0.045 GHz for variations in any of the following independent variables.

   a) Eb = Eb $\pm$ 1%, Po(2), Po = 1KW

   b) F = 9.200 $\pm$ .045 GHz, Po(2), Po = 700W

9. The tube shall operate satisfactorily from sea level 10,000 feet.

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10. The tube shall not oscillate or be damaged when operated into a load VSWR of 3:1 at any phase angle.

11. The second harmonic output shall be at least 30db below the output power level [Po(1)]. All spurious outputs excluding harmonics shall be at least 120 db below the output power level from 5GHz to 18GHz.

12. Two minutes after the application of the heater voltage to a tube stabilized at room temperature, the full Power Output test condition shall be applied. The tube shall pass the Power Output test in 3 seconds after the application of the full test condition voltages and rf drive power.

13. The tube shall be considered to have failed this test by inability to comply with the specified life test end points.

14. The tube shall operate satisfactorily during and after being subjected to the extreme temperatures of -40°C (-40°F) to +79.4°C (+175°F).

15. The tube shall be vibrated sinusoidally along three mutually perpendicular axes, one of which shall be normal to the mounting surface of the tube. The tube shall operate during this test and meet the requirements of Q.C.I.1 after this test. Vibration shall be in accordance with the following schedule:

	Amplitude 1/2 Total Excursion	Acceleration	Time (Each Axis)
0 - 10 Hz	0.060 ich	---	1 Hour
11 - 20 Hz	0.030 inch	---	2 hours
21 - 400 Hz	---	9G	2 hours

16. The tube shall be shocked (6 impact shocks) in each of 3 mutually perpendicular axes, one of which shall be parallel to the beam axis. A total of 18 impact shocks, 3 in each direction, shall be administered. The tube shall operate during this test and meet the requirements of Q.C.I.1 after this test.

17. The tube shall be shocked on (3 total) in each of three mutually perpendicular axes. Upon completion of this test, the tube shall meet the requirements of this specification (Q.C.I.1).

18. This measurement shall be made in air between the ends of the tube electrode leads.

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19. Tuner Life - The tube shall be capable of being set to any randomly selected channel referenced in note 6, without impairing the operation of the tube for any one of the following parameters:

    1) Over the 100 MHz frequency range (10548235) 100 settings

    2) Over the 50 MHz frequency range (10549169) 250 settings

    3) Over the 25 MHz frequency range (10549169) 1000 settings

20. The 3 db instantaneous bandwidth shall be 5 MHz minimum (Linear Mode of Operation) for any frequency in the band when Po = 500W.

21. The tube shall exhibit a linear mode of operation if it complies with the following: As Pd is decreased by 9 db below that Pd established at Po = 1 KW, the differential gain shall be less than .6 db. Diffential gain is defined as the difference in gain, obtained by comparing the decrease in the power output to the 9 db decrease in drive power.

22. Coarse and Fine Tuning requirements (Note 7) shall not apply during saturated drive conditions.

23. Saturated Po(1) and Gain Po(1)

    For purposes of this specification, the value of saturated peak power and gain referred to as Po(1), shall be determined in the following manner:

    Step 1. Adjust the test frequency to the specified value.

    Step 2. Adjust Pd and sync tune cavities #1, #2, #3 and #5 for Po(2) Po = 700W.

    Step 3. Increase Pd to peak Po max.

    Step 4. Retune #5 cavity for Po max.*

    Step 5. Record Po average and pulse width.

    Step 6. Determine duty ratio and compute peak power and tube gain.

    *Do not readjust Pd after this step.

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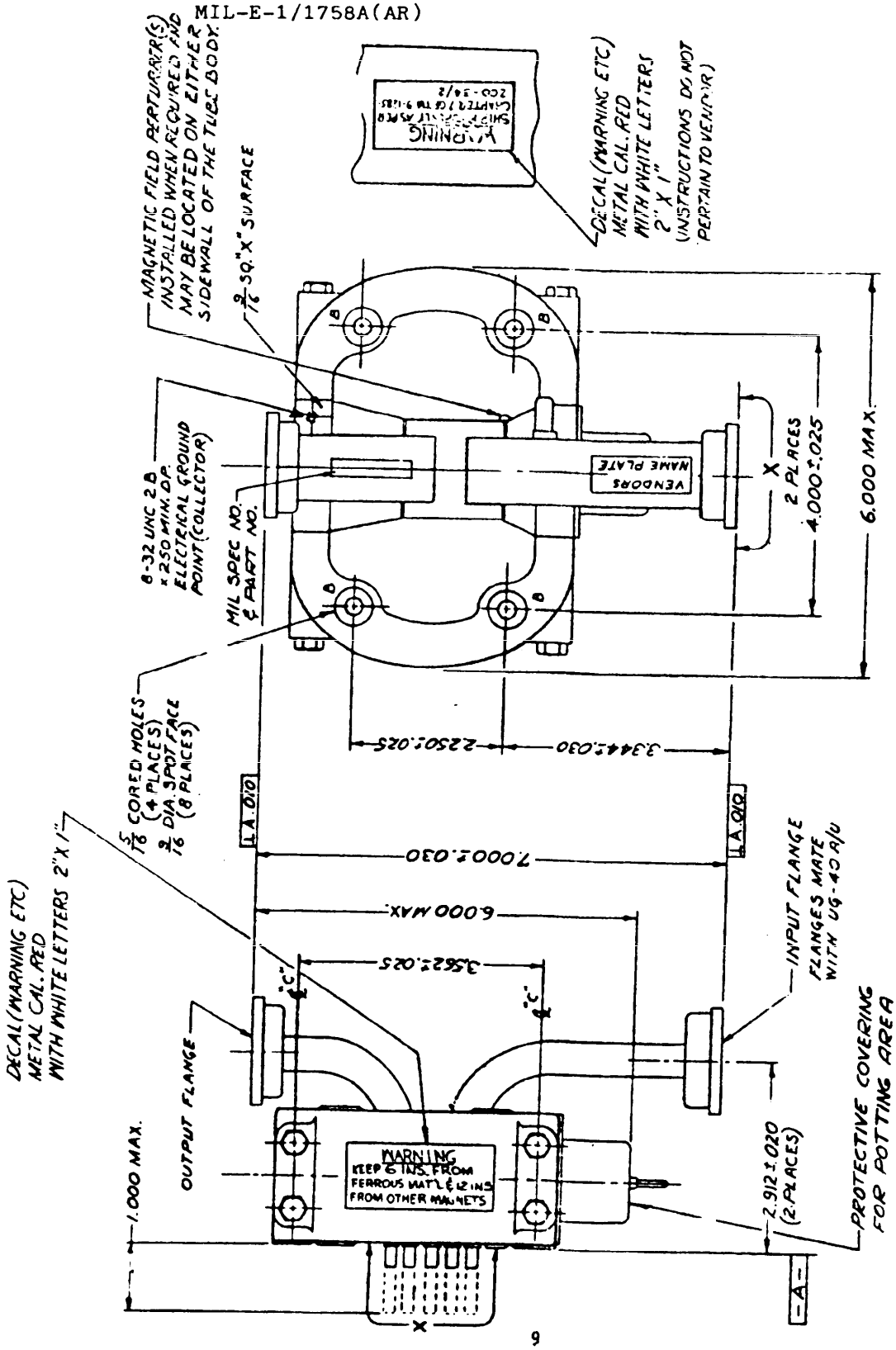
24. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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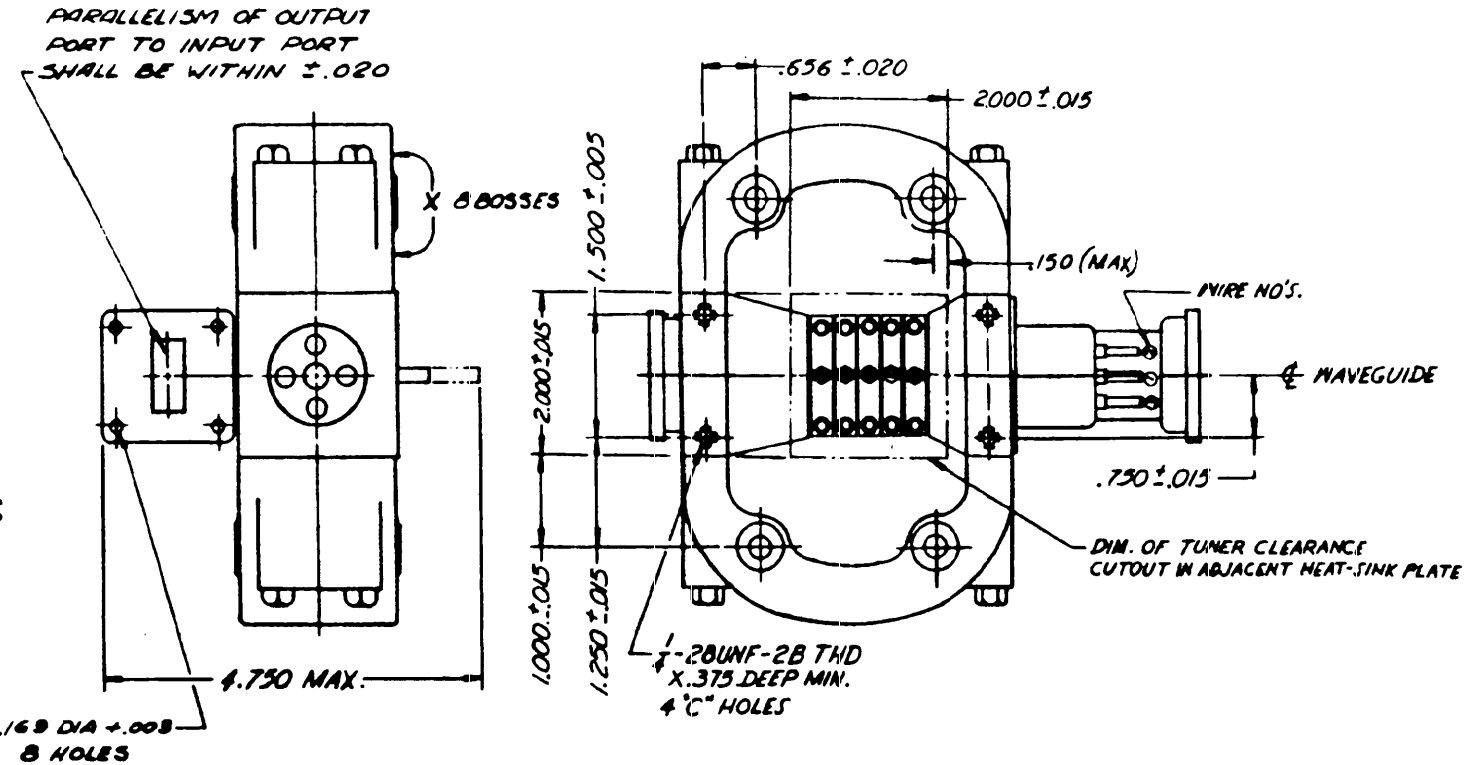
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KLYSTRON POWER AMPLIFIER

FIGURE 1 SHEET 1

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WIRE NO.	WIRE LENGTH	COLOR CODE	APPLIED VOLTAGES
1		YELLOW = HEATER	0.0VDC
2		WHITE = HEATER - CATHODE	+5.0VDC
3		RED = MOD ELECTRODE	

ALL LEAD LENGTHS 18" NOMINAL, 17" MINIMUM

KLYSTRON POWER AMPLIFIER

FIGURE 1 SHEET 2

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